THE KERNEL OF THE FIRST ORBIT HOMOMORPHISM ON THE POSITIVE WITT ALGEBRA, VIA THE CONLEY–GOODE METHOD

ABSTRACT. Let Ψ be the first orbit homomorphism of Pham–Timmins, restricted to the enveloping algebra of the positive Witt algebra $W_{\geq 1}$. We prove, with the Conley–Goode criterion, that $\ker \Psi$ is the two-sided ideal generated by the single differentiator $\Omega = \Omega^{(3)}_{5,1}$, recovering a result of Sierra–Walton. The proof is self-contained and uses only elements of positive weight. An exact formula for Ψ shows that Ψ of a two-letter word is quadratic in its index. This gives, in turn: injectivity on normal forms; the quadratic kernel as the span of third differences; and the Conley–Goode condition on quadratic elements, by Lagrange interpolation. Exact bracket formulas for differentiators then show that Ω generates them all.

1. STATEMENT

Throughout, $\Bbbk$ is a field of characteristic 0.

The Lie algebra. Let $\mathfrak{g} = W_{\geq 1}$ have basis $\{e_i : i \geq 1\}$ and bracket $[e_i, e_j] = (j-i)\, e_{i+j}$. Let $U = U(\mathfrak{g})$, filtered by order: U_m is spanned by products of at most m elements of $\mathfrak{g}$. By PBW, the nondecreasing monomials $e_{i_1} \cdots e_{i_m}$ form a basis of U, compatible with the filtration, and $S(\mathfrak{g}) = \operatorname{gr} U = \Bbbk[\bar{e}_1, \bar{e}_2, \dots]$. For $z \in U_m$ write $\operatorname{gr}_m z = z + U_{m-1}$, and for a subspace $A \subseteq U$ put $\operatorname{gr}_m A = (A \cap U_m + U_{m-1})/U_{m-1}$. The map gr is multiplicative, and the symmetrizer $\operatorname{sym} : S(\mathfrak{g}) \to U$ satisfies $\operatorname{gr}_m \operatorname{sym}(x) = x$ for $x \in S^m(\mathfrak{g})$.

Each $\operatorname{ad} e_l$ preserves every U_m, and on $S(\mathfrak{g})$ it induces the derivation θ_l with $\theta_l(\bar{e}_a) = (a-l)\, \bar{e}_{a+l}$; that is, $\operatorname{gr}_m [e_l, z] = \theta_l(\operatorname{gr}_m z)$ for $z \in U_m$.

Setting $\deg e_i = i$ (the *weight*) grades $\mathfrak{g}$, U and $S(\mathfrak{g})$, since $[e_i, e_j] \in \Bbbk e_{i+j}$. No element e_0 is needed.

The target. Let $T = A_1[v] = \Bbbk[t, \partial] \otimes \Bbbk[v]$, with $\partial t - t\partial = 1$ and v central, graded by weight $\deg t = 1$, $\deg \partial = -1$, $\deg v = 0$.

Theorem 1.1. *There is an algebra homomorphism $\Psi : U \to T$ with $\Psi(e_i) = t^{i+1}\partial + (i+1)t^i v$, and*

$$\ker \Psi = (\Omega), \qquad \Omega = e_5 e_1 - 3e_4 e_2 + 3e_3 e_3 - e_2 e_4 = e_1 e_5 - 4e_2 e_4 + 3e_3^2 + 2e_6.$$

Here Ψ is the restriction of the first orbit homomorphism Ψ_1 of [2], and $\Omega = \Omega^{(3)}_{5,1}$ in their notation.

2. THE HOMOMORPHISM IN GOOD COORDINATES

Put $X = t\partial + 2v \in T$.

Lemma 2.1. (i) *X commutes with v, and $Xt = t(X+1)$. Hence $g\, t^j = t^j\, g(X+j)$ for every $g \in \Bbbk[X, v]$, where $g(X+j)$ means $g(X+j, v)$.*

(ii) *X and v are algebraically independent, and if $t^w g = 0$ with $g \in \Bbbk[X, v]$ then $g = 0$.*

Proof. (i) follows from $(t\partial)t = t(t\partial + 1)$. For (ii), note that $\Bbbk[X, v] = \Bbbk[t\partial, v]$, and $(t\partial)^a = t^a \partial^a + (\text{a combination of } t^c \partial^c,\ c < a)$. So the elements $(t\partial)^a v^c$ are linearly independent in T, which has basis $t^a \partial^b v^c$. Finally, T is a domain and $t^w \neq 0$. □

For $i \geq 1$ let $\ell_i = X + (i-1)v$ and $Y_i = t^i \ell_i = t^{i+1}\partial + (i+1)t^i v$.

Lemma 2.2. *For $i_1, \dots, i_m \geq 1$ with $i_1 + \cdots + i_m = w$,*

$$Y_{i_1} \cdots Y_{i_m} = t^w \prod_{k=1}^{m} \ell_{i_k}(X + \sigma_k), \qquad \sigma_k = i_{k+1} + \cdots + i_m. \tag{1}$$

In particular $[Y_i, Y_j] = (j-i)\, Y_{i+j}$, so Ψ exists, with $\Psi(e_i) = Y_i$.

Proof. Move all powers of t to the left using Lemma 2.1(i). The k-th factor ℓ_{i_k} then has t^{σ_k} passing it, so it is shifted by σ_k. For $m = 2$ this gives

$$Y_iY_j - Y_jY_i = t^{i+j}\Big[\big(X+j+(i-1)v\big)\big(X+(j-1)v\big) - \big(X+i+(j-1)v\big)\big(X+(i-1)v\big)\Big] = (j-i)\, t^{i+j}\ell_{i+j}.$$

□

So for a word $n = e_{i_1} \cdots e_{i_m}$ of weight w we have $\Psi(n) = t^w f_n$, where $f_n = \prod_k \ell_{i_k}(X + \sigma_k) \in \Bbbk[X, v]$.

Corollary 2.3. $\Psi(\Omega) = 0$.

Proof. For $a + b = 6$ we have $\Psi(e_a e_b) = t^6 F(b)$, where $F(b) = \big(X + b + (5-b)v\big)\big(X + (b-1)v\big)$ is a polynomial of degree 2 in b. Hence

$$\Psi(\Omega) = t^6\big(F(1) - 3F(2) + 3F(3) - F(4)\big) = 0,$$

since the third finite difference of a quadratic polynomial vanishes. □

3. The Conley–Goode criterion

Theorem 3.1 ([1, Prop. 3.1])**.** *Let $\Psi : U(\mathfrak{g}) \to T$ be an algebra map and $d \geq 1$. Let $J \subseteq \mathfrak{g}$, $\mathcal{J} \subseteq U(\mathfrak{g})$ and $I \subseteq \ker \Psi \cap U_d$ be subspaces such that:*

(a) *Ψ is injective on $\mathcal{J}$;*
(b) *$U_{d-1} \subseteq \mathcal{J}$;*
(c) *$\operatorname{gr}_m \mathcal{J} = S^{m-d+1}(J)\, S^{d-1}(\mathfrak{g})$ for all $m \geq d$;*
(d) *$\operatorname{proj}_J(\operatorname{gr}_d I) = S^d(\mathfrak{g}/J)$, where $\operatorname{proj}_J : S(\mathfrak{g}) \to S(\mathfrak{g}/J)$ is induced by $\mathfrak{g} \to \mathfrak{g}/J$.*

Then $\ker \Psi = (I)$.

Proof. Choose $\mathfrak{g} = J \oplus C$. Call the number of factors from C in a monomial its *C-degree*. The kernel of the algebra map proj_J is the ideal generated by J, so $\ker \operatorname{proj}_J \cap S^d(\mathfrak{g}) = J\, S^{d-1}(\mathfrak{g})$.

Step 1: $S^m(\mathfrak{g}) = \operatorname{gr}_m \mathcal{J} + S^{m-d}(\mathfrak{g})\, \operatorname{gr}_d I$ for $m \geq d$. Let M be a monomial of order m and C-degree c, and induct on c. If $c < d$, then $M \in \operatorname{gr}_m \mathcal{J}$ by (c). Otherwise write $M = rq$, with q a product of d factors from C. By (d), choose $p \in \operatorname{gr}_d I$ with $\operatorname{proj}_J(p) = \operatorname{proj}_J(q)$. Then $q - p \in J\, S^{d-1}(\mathfrak{g})$, so in $M = rp + r(q-p)$ the second term has C-degree less than c.

Step 2: $U = \mathcal{J} + (I)$. We show $U_m \subseteq \mathcal{J} + (I)$ by induction on m; the base case $m < d$ is (b). Let $x \in U_m$ with $m \geq d$. By Step 1, $\operatorname{gr}_m x = \operatorname{gr}_m y + \sum_k r_k \operatorname{gr}_d z_k$ with $y \in \mathcal{J}$, $r_k \in S^{m-d}(\mathfrak{g})$ and $z_k \in I$. Then $x - y - \sum_k \operatorname{sym}(r_k) z_k \in U_{m-1}$.

Step 3. Clearly $(I) \subseteq \ker \Psi$. Conversely, if $z \in \ker \Psi$, write $z = y + i$ with $y \in \mathcal{J}$ and $i \in (I)$. Then $\Psi(y) = 0$, so $y = 0$ by (a). □

4. The data, and conditions (b) and (c)

We take $d = 2$ and $J = \operatorname{span}\{e_1, e_2\}$. Let $\mathcal{N}$ be the set of PBW monomials

$$e_1^a e_2^b \qquad \text{and} \qquad e_1^a e_2^b e_j \quad (j \geq 3),$$

that is, polynomials in e_1, e_2 times at most one higher generator, and let $\mathcal{J} = \operatorname{span} \mathcal{N}$. For I we take the whole quadratic part of the kernel,

$$I = K := \ker \Psi \cap U_2.$$

Since $\mathcal{N}$ is part of the PBW basis, $\operatorname{gr}_m \mathcal{J}$ is spanned by the symbols of the elements of $\mathcal{N}$ of length m. So

$$\operatorname{gr} \mathcal{J} = \Bbbk[\bar{e}_1, \bar{e}_2] \ \oplus \ \bigoplus\nolimits_{j\geq 3} \bar{e}_j \, \Bbbk[\bar{e}_1, \bar{e}_2].$$

Condition (b) holds because $1, e_1, e_2, e_j \in \mathcal{N}$. For (c), $S^{m-1}(J)\, S^1(\mathfrak{g})$ is spanned by the monomials $\bar{e}_1^a \bar{e}_2^b \bar{e}_j$ with $a+b = m-1$ and $j \geq 1$. Those with $j \leq 2$ are exactly the $\bar{e}_1^a \bar{e}_2^b$ with $a+b=m$, so this span is $\operatorname{gr}_m \mathcal{J}$.

5. Condition (a)

Proposition 5.1. *Ψ is injective on $\mathcal{J}$.*

Proof. Ψ preserves weight, T is the direct sum of its weight spaces, and $\mathcal{N}$ consists of weight vectors. By Lemma 2.1(ii) it therefore suffices to show, for each w, that the polynomials f_n with $n \in \mathcal{N}$ of weight w are linearly independent in $\Bbbk[X, v]$. By (1),

$$f_{e_1^a e_2^b e_j} = \prod_{p=0}^{a-1} (X + w - a + p) \cdot \prod_{q=0}^{b-1} (X + j + 2q + v) \cdot \big(X + (j-1)v\big),$$

$$f_{e_1^a e_2^b} = \prod_{p=0}^{a-1} (X + w - a + p) \cdot \prod_{q=0}^{b-1} (X + 2q + v).$$

Order the monomials of $\Bbbk[X, v]$ lexicographically with $v > X$. The leading monomial of $X + c$ is X. That of $X + c + v$ is v, and so is that of $X + (j-1)v$, because $j \neq 1$. Leading monomials multiply, so

$$\operatorname{lm} f_{e_1^a e_2^b e_j} = X^a v^{b+1}, \qquad \operatorname{lm} f_{e_1^a e_2^b} = X^a v^b.$$

For fixed w these are pairwise distinct. A monomial $X^a v^k$ is the leading monomial of at most one mixed normal form, namely the one with $b = k-1$ and $j = w - a - 2b$. It is also the leading monomial of at most one pure normal form, namely the one with $b = k$. It cannot come from both, because then $j = w - a - 2(k-1) = 2$. Polynomials with distinct leading monomials are linearly independent. □

6. The quadratic kernel

For $k \geq 4$ and $s \geq 1$ put $c_i = (-1)^i \binom{3}{i}$ (with $c_i = 0$ for $i \notin \{0, 1, 2, 3\}$) and

$$D_{k,s} = \sum_i c_i \, e_{k-i} e_{s+i} = e_k e_s - 3 e_{k-1} e_{s+1} + 3 e_{k-2} e_{s+2} - e_{k-3} e_{s+3},$$

a combination of words of weight $k + s$. So $D_{k,s} = \Omega^{(3)}_{k,s}$, and $\Omega = D_{5,1}$.

Lemma 6.1. *Fix $w \geq 3$. For $1 \leq c \leq w-1$ we have $\Psi(e_{w-c} e_c) = t^w F(c)$, where*

$$F(c) = F_0 + c\, F_1 + c^2 F_2, \qquad F_0 = \big(X + (w-1)v\big)(X - v), \quad F_1 = X - v + wv^2, \quad F_2 = v - v^2,$$

and F_0, F_1, F_2 are linearly independent in $\Bbbk[X, v]$.

Proof. By (1), $F(c) = \big(X + c + (w - c - 1)v\big)\big(X + (c-1)v\big)$. Expanding in powers of c gives the stated F_0, F_1, F_2. Only F_0 contains X^2, and only F_1 contains X without v, while $F_2 \neq 0$. So they are independent. □

Proposition 6.2. *Let $K = \ker \Psi \cap U_2$. For every w, the weight-w part K_w is spanned by the $D_{w-s,s}$ with $1 \leq s \leq w-4$. In particular every $D_{k,s}$ lies in $\ker \Psi$.*

Proof. For $w \leq 2$ both sides are 0, since Ψ is injective on $U_1 \subseteq \mathcal{J}$. Let $w \geq 3$. Every element of $(U_2)_w$ is a combination of words $e_{w-c}e_c$, because $e_w = (e_{w-1}e_1 - e_1e_{w-1})/(2-w)$. Consider the linear map

$$\lambda = (\lambda_1, \dots, \lambda_{w-1}) \in \Bbbk^{w-1} \;\longmapsto\; z_\lambda = \sum_c \lambda_c \, e_{w-c}e_c \in (U_2)_w,$$

which is onto. By Lemma 6.1,

$$\Psi(z_\lambda) = t^w\Big(F_0 \sum_c \lambda_c + F_1 \sum_c \lambda_c c + F_2 \sum_c \lambda_c c^2\Big).$$

So $z_\lambda \in \ker \Psi$ if and only if λ lies in

$$V = \Big\{\lambda : \; \textstyle\sum_c \lambda_c \, c^r = 0 \;\; \text{for } r = 0, 1, 2\Big\}.$$

The three conditions are independent (Vandermonde, as $w - 1 \geq 3$), so $\dim V = w - 4$. For $1 \leq s \leq w - 4$, let $\delta^{(s)}$ have entries c_0, c_1, c_2, c_3 in positions $s, \dots, s + 3$ and 0 elsewhere. Then $\delta^{(s)} \in V$, because a third finite difference kills polynomials of degree ≤ 2. The $\delta^{(s)}$ are independent, since $\delta^{(s)}$ is the first one with a nonzero entry in position s. Hence $V = \operatorname{span}\{\delta^{(s)}\}$. Finally $z_{\delta^{(s)}} = D_{w-s,s}$, so $K_w = z(V) = \operatorname{span}\{D_{w-s,s}\}$. □

7. Condition (d)

Proposition 7.1. $\operatorname{proj}_J(\operatorname{gr}_2 K) = S^2(\mathfrak{g}/J)$.

Proof. $S^2(\mathfrak{g}/J)$ is spanned by the classes of the products $\bar{e}_a\bar{e}_b$ with $3 \leq a \leq b$. Fix such a, b and put $w = a + b \geq 6$. By Lemma 6.1, F is a polynomial of degree 2, and the nodes $1, 2, w - 1$ are distinct. So Lagrange interpolation gives $F(b) = \sum_{c\in\{1,2,w-1\}} L_c(b)F(c)$, where L_c are the Lagrange basis polynomials. Hence

$$z = e_ae_b - L_1(b)\, e_{w-1}e_1 - L_2(b)\, e_{w-2}e_2 - L_{w-1}(b)\, e_1e_{w-1} \;\in K.$$

Each subtracted word contains e_1 or e_2, so $\operatorname{gr}_2 z \equiv \bar{e}_a\bar{e}_b$ modulo $J\,S^1(\mathfrak{g})$, and $\operatorname{proj}_J(\operatorname{gr}_2 z)$ is the class of $\bar{e}_a\bar{e}_b$. □

Conditions (a)–(d) hold by Proposition 5.1, Section 4 and Proposition 7.1, with $I = K \subseteq \ker \Psi \cap U_2$. Theorem 3.1 therefore gives

$$\ker \Psi = (K). \tag{2}$$

8. The differentiators are generated by Ω

Lemma 8.1. *For $k \geq 4$ and $s \geq 1$:*

(i) $D_{k,s} = -D_{s+3,k-3}$. *In particular* $D_{4,1} = D_{5,2} = 0$ *and* $D_{4,2} = -D_{5,1}$.
(ii) $[e_1, D_{k,s}] = (k-1)\, D_{k+1,s} + (s+2)\, D_{k,s+1}$.
(iii) $[e_2, D_{k,s}] = (k-2)\, D_{k+2,s} + 3\, D_{k+1,s+1} + (s+1)\, D_{k,s+2}$.

Proof. (i) Reindexing $i \mapsto 3 - i$ gives $D_{s+3,k-3} = -\sum_i c_i \, e_{s+i}e_{k-i}$. Hence

$$D_{k,s} + D_{s+3,k-3} = \sum_i c_i \, [e_{k-i}, e_{s+i}] = \Big(\sum_i c_i \, (s - k + 2i)\Big) e_{k+s} = 0,$$

since $\sum_i c_i = \sum_i c_i \, i = 0$.

(ii) and (iii). For words, $[e_l, e_ae_b] = (a-l)\, e_{a+l}e_b + (b-l)\, e_ae_{b+l}$ exactly. So both sides of (ii) are combinations of the words $e_{k+1-r}e_{s+r}$, and comparing coefficients, (ii) amounts to

$$c_r(k-r-1) + c_{r-1}(s+r-2) = (k-1)c_r + (s+2)c_{r-1}, \quad \text{i.e.} \quad r\,c_r = (r-4)\, c_{r-1} \qquad (\text{all } r).$$

This holds because $\binom{3}{r}/\binom{3}{r-1} = (4-r)/r$. Similarly, comparing coefficients of $e_{k+2-r}e_{s+r}$, (iii) amounts to

$$r\,c_r + 3\,c_{r-1} = (r-5)\,c_{r-2} \qquad (\text{all } r),$$

which one checks for $r = 0, \dots, 5$, using $(c_0, \dots, c_3) = (1, -3, 3, -1)$. □

Proposition 8.2. *Let M be the $\mathrm{ad}(\mathfrak{g})$-submodule of U generated by Ω. Then $D_{k,s} \in M$ for all $k \geq 4$ and $s \geq 1$.*

Proof. We induct on the weight $w = k + s$. The weight-w differentiators are $D_{w-s,s}$ with $1 \leq s \leq w-4$.

Weights ≤ 7. By Lemma 8.1(i), $D_{4,1} = 0$, $D_{4,2} = -\Omega$ and $D_{5,2} = 0$. By (ii), $[e_1, \Omega] = 4D_{6,1} + 3D_{5,2} = 4D_{6,1}$, so $D_{6,1} \in M$, and then $D_{4,3} = -D_{6,1} \in M$.

From weights $w-1, w$ to $w+1$, for $w \geq 7$. Put $X_0 = D_{w,1}$. For $1 \leq s \leq w-4$, (ii) gives

$$[e_1, D_{w-s,s}] = (w-s-1)\,D_{w+1-s,s} + (s+2)\,D_{w-s,s+1} \in M.$$

By induction on s, $D_{w+1-s,s} \equiv \mu_s\,X_0 \pmod{M}$ for $1 \leq s \leq w-3$, where $\mu_1 = 1$ and $\mu_{s+1} = -\frac{w-s-1}{s+2}\,\mu_s$. In particular $\mu_2 = -\frac{w-2}{3}$ and $\mu_3 = \frac{(w-2)(w-3)}{12}$. Now (iii) applied to $D_{w-2,1} \in M$ gives

$$\begin{aligned} M \ni [e_2, D_{w-2,1}] &= (w-4)\,X_0 + 3\,D_{w-1,2} + 2\,D_{w-2,3} \\ &\equiv \big(w - 4 + 3\mu_2 + 2\mu_3\big)X_0 = \frac{(w-6)(w+1)}{6}\,X_0. \end{aligned}$$

The coefficient is nonzero for $w \geq 7$, so $X_0 \in M$, and therefore every $D_{w+1-s,s} \in M$. □

9. Proof of Theorem 1.1

By (2), Proposition 6.2 and Proposition 8.2,

$$\ker\Psi = (K) = \big(D_{k,s} : k \geq 4,\ s \geq 1\big) \subseteq (M) \subseteq (\Omega) \subseteq \ker\Psi,$$

using $[e_l, z] = e_l z - z e_l$ for $M \subseteq (\Omega)$, and Corollary 2.3 for the last inclusion. Hence $\ker\Psi = (\Omega)$. □

Remark 9.1.

(1) The interpolation argument of Proposition 7.1 also works in Pham–Timmins' setting $U(W_{\geq -1})$, with $J = \mathrm{span}\{e_{-1}, \dots, e_{n-1}\}$ and $I = \ker\Psi_n \cap U_2$. There $\Psi_n(e_{w-c}e_c)$ is a polynomial in c of degree at most $2n$, and the $2n+2$ words with a letter in J supply enough interpolation nodes. (We checked the resulting statement by machine for $n \leq 3$ and weights ≤ 14.)

(2) Lemma 8.1, Proposition 6.2 (in weights ≤ 14) and the elements z of Proposition 7.1 were also confirmed by machine.

References

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